

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data to Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a PySpark job to read customer and order CSV data and load it into delta tables.

Target Cluster Configuration:

- Cluster Name: Gen AI POC Databricks Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, spark-csv

Source Data Details:

- Dataset Name: Customer Data
- Location: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata``
- Format: CSV
- Description: Customer data in CSV format
- Dataset Name: Order Data
- Location: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``
- Format: CSV
- Description: Order data in CSV format

Target Data Details:

- Output Name: Customer Delta Table
- Location: ``gen_ai_poc_databrickscoe.sdlc_wizard.customer``
- Format: Delta
- Description: Customer data in delta format
- Output Name: Order Delta Table

- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`
- Format: Delta
- Description: Order data in delta format

Job Flow / Pipeline Stages:

- Read customer CSV data and load it into a delta table named "customer".
- Read order CSV data and load it into a delta table named "order".
- Add a new column "TotalAmount" to the "order" delta table.

Data Transformations / Business Logic:

- Step: Read customer CSV data
- Description: Read customer CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata`
- Transformation Logic: Use `spark.read.csv` to read CSV data
- Step: Load customer data into delta table
- Description: Load customer data into a delta table named "customer"
- Transformation Logic: Use `deltaTable.write` to write data to delta table

Error Handling and Logging:

- Log errors to a logging table or file
- Handle null or empty input data

TR-DLT-002: Clean and Process Customer and Order Data

Related Functional Requirement(s): FR-DLT-002

Objective: Implement a PySpark job to remove null and duplicate records from customer and order delta tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: Customer Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Customer data in delta format
- Dataset Name: Order Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`
- Format: Delta
- Description: Order data in delta format

Target Data Details:

- Output Name: Cleaned Customer Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Cleaned customer data in delta format
- Output Name: Cleaned Order Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`
- Format: Delta
- Description: Cleaned order data in delta format

Job Flow / Pipeline Stages:

- Remove null records from both "customer" and "order" delta tables.
- Remove duplicate records from both "customer" and "order" delta tables.

Data Transformations / Business Logic:

- Step: Remove null records
- Description: Remove null records from customer and order delta tables
- Transformation Logic: Use `filter` to remove null records

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-003: Create and Load ordersummary Table

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a PySpark job to create an "ordersummary" table and load it with joined customer and order data using SCD type 2.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: Customer Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Customer data in delta format
- Dataset Name: Order Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`
- Format: Delta
- Description: Order data in delta format

Target Data Details:

- Output Name: ordersummary Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.ordersummary`

- Format: Delta
- Description: Joined customer and order data in delta format

Job Flow / Pipeline Stages:

- Create "ordersummary" table if it does not exist.
- Join "customer" and "order" delta tables using "CustId" field.
- Load the joined data into "ordersummary" table using SCD type 2.

Data Transformations / Business Logic:

- Step: Join customer and order data
- Description: Join customer and order delta tables using "CustId" field
- Transformation Logic: Use `join` to join data

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-004: Update ordersummary Table on Customer Changes

Related Functional Requirement(s): FR-DLT-004

Objective: Implement a PySpark job to update the "ordersummary" table whenever there is a change in the "customer" table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: Customer Delta Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Customer data in delta format

Target Data Details:

- Output Name: Updated ordersummary Table
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.ordersummary`
- Format: Delta
- Description: Updated joined customer and order data in delta format

Job Flow / Pipeline Stages:

- Detect changes in the "customer" table.
- Update the "ordersummary" table by making old records inactive and new records active.
- Update StartDate and EndDate accordingly to maintain history.

Data Transformations / Business Logic:

- Step: Detect changes in customer data
- Description: Detect changes in customer delta table

- Transformation Logic: Use `exceptAll` to detect changes

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-005: Create and Load customeraggregatespend Table

Related Functional Requirement(s): FR-DLT-005

Objective: Implement a PySpark job to create a "customeraggregatespend" table and load it with aggregated data from "ordersummary" table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: ordersummary Table
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.ordersummary`
- Format: Delta
- Description: Joined customer and order data in delta format

Target Data Details:

- Output Name: customeraggregatespend Table
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.customeraggregatespend`
- Format: Delta
- Description: Aggregated customer spend data in delta format

Job Flow / Pipeline Stages:

- Create "customeraggregatespend" table if it does not exist.
- Aggregate "TotalAmount" column from "ordersummary" table and group by "Name" and "Date" columns.
- Load the aggregated data into "customeraggregatespend" table.

Data Transformations / Business Logic:

- Step: Aggregate customer spend data
- Description: Aggregate "TotalAmount" column from "ordersummary" table
- Transformation Logic: Use `groupBy` and `sum` to aggregate data

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-006: Implement Delta Live Tables

Related Functional Requirement(s): FR-DLT-006

Objective: Implement delta live tables for the entire workflow.

Target Cluster Configuration: Same as TR-DLT-001

Job Flow / Pipeline Stages:

- Implement delta live tables for loading customer and order data.
- Implement delta live tables for cleaning and processing customer and order data.
- Implement delta live tables for creating and loading "ordersummary" table.
- Implement delta live tables for updating "ordersummary" table on customer changes.
- Implement delta live tables for creating and loading "customeraggregatespend" table.

Data Transformations / Business Logic: Same as previous TRs

Error Handling and Logging: Same as TR-DLT-001